

Science 24

UNIT B WORKBOOK



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Chapter 5: Energy Transformation

Define the following:

Energy

Potential Energy

Kinetic Energy

Solar Energy

Thermal Energy

Wind Energy

Geothermal Energy

Input Energy

Converter

Output Energy

Chemical Energy

1. Law of Conservation of Energy

2. List 3 examples of kinetic energy. (pg. 82)

3. List 3 examples of potential energy. (pg. 82)

4. List the energies you depend on when you are: (pgs. 82, 84 & 85)

a. Walking to school

b. Driving to the store

c. Turning on the computer

d. Listening to the radio

5. List two ways the Sun's energy is converted for use. (pgs. 84-85)

6. Is thermal energy a form of kinetic energy? Explain. (pg. 84)

7. What energy conversions take place from the eating of an apple to the energy used to throw a ball? (pgs. 84-85)

8. Briefly explain how plants and animals use energy from the Sun. (pg. 84)

9. List 8 forms of energy and give 2 examples of each. (pg. 85)

10. What 3 components are necessary for an energy conversion? (pg 86)

11. Identify the energy conversions that take place in the following scenarios.

a. When the inside of a car heats up when the car is left in the Sun.

b. When you paddle a canoe

c. A model rocket taking off

d. Drill press cutting a piece of wood

12. List 3 examples of energy conversion in which heat is given off. (pgs. 86 to 91)

13. Give two examples of waste thermal energy that might cause a problem? (pg. 91)

14. List 3 kinds of energy that come from a blow dryer. (pg. 92)

15. Most of the energy produced by a car engine is not used to make the car move.
Where does the rest of the energy go? (pg. 92)

16. Use an example to explain the Law of Conservation of Energy. (pg. 92)

17. Explain why input energy is greater than useful output energy. (pg. 94)

18. Sketch two machines that have not been discussed but you are familiar with.
On the sketch, identify where there is waste output energy. (pg. 94)

Chapter 6:Generating Electric Energy

Define the following:

Generator

Turbine

Hydroelectric

Thermoelectric

Thermonuclear

Solar Cells

Newton

Power

Watt

Watt Hour (W*H)

Kilowatt Hours (kW*h)

Efficiency

4. List 3 ways energy is generated in Canada. (pg. 102)

5. Describe how coal is used to produce electricity. (pg. 104)

6. List two advantages of generating electric energy using a hydroelectric dam. (pg. 105)

7. What are some possible ecological damages that may result because of a hydroelectric dam? (pg. 105)

8. How much energy is lost as electricity moves from generating stations to your home?
How is it lost? (pg. 106)

9. List 3 steps that take place as nuclear energy is converted to electric energy. (pg. 106)

10. What does a thermo-electric plant have in common with a thermonuclear power plant? (pg. 106)

11. What is power? (pg. 108)

12. What are the units used to measure the power of electric appliances?(pgs. 109 & 110)

13. Calculate the following examples using the formula
(REMEMBER time should be in seconds).

$$\text{Power} = \frac{\text{energy}}{\text{time}}$$

a. 60 joules of energy are used in 2 hours

b. 120 joules in half a minute

c. 200 joules in 50 seconds

14. What are the units for the energy we buy from the power company. (pg. 110)

15. Using the formula $\text{energy} = \text{power} * \text{time}$ calculate the following examples.

a. A 100W light bulb is left on for 4 hours.

b. A 240 Watt fan is left on for 2 hours.

c. A 540 W stove is left on for 3 hours.

16. What is the difference between a kWh and a Wh? (pgs. 109 & 110)

17. Why would the electric company use kilowatt hours instead of watt hours? (pg. 110)

18. How do we calculate efficiency? (pg. 112)

19. Why does an incandescent light bulb get so much hotter than a fluorescent light bulb? (pg. 111)

20. Use an example to define "efficiency". (pgs. 111 to 113)

21. State 3 ways to conserve energy at home. (pg. 113)

Chapter 7: Energy for Life

Define the following:

Chloroplasts

Chlorophyll

Photosynthesis

Cellular Respiration

Homeostasis

Metabolism

Carbohydrate

Glycogen

Fat

Saturated Fats

Unsaturated Fats

Protein

Metabolic needs

1. Where in plants does photosynthesis take place? (pg. 118)
2. Write the formula for photosynthesis. (pg. 118)
3. Where does cellular respiration take place? (pg. 119)
4. Write the formula for cellular respiration. (pg. 119)
5. What is the main source of energy of all the cells in our body? (pg. 119)
6. What would happen if we had no mitochondria in our bodies? (pg. 119)

7. Why is cellular respiration important? (pg. 119)

8. What does lactic acid do to your muscles? (pg. 119)

9. How do animals capture energy from the sun? (pg. 122)

10. As you go higher on the pyramid of energy, is there more or less energy available?
Why is this so? (pg. 122)

11. Why are there less organisms as you move up a food chain? (pg. 122)

12. How much energy is transferred to the next level in a food chain? Pg 122
13. Draw a food chain that is at least 4 animals long. Label the Producer, Primary consumer, Secondary consumer and the tertiary consumer . (pg. 122)
14. What is the internal balance in the body called? (pg. 123)
15. List 2 things that your body will do to cool itself down? (pg. 123)
16. List 2 things your body does to warm itself up? (pg. 123)
17. List 2 adaptations that the Inuit have to keep them warm. (Notes)

18. List 2 adaptations that desert people have that allows them to survive. (Notes)

19. How does homeostasis and metabolism work together to keep an organism alive?
(pg. 124)

20. What is the main source of energy for the human body? (pg. 125)

21. What role do carbohydrates play in cellular respiration? (pg.125)

22. What percent of our daily calories should be carbohydrates? (pg. 125)

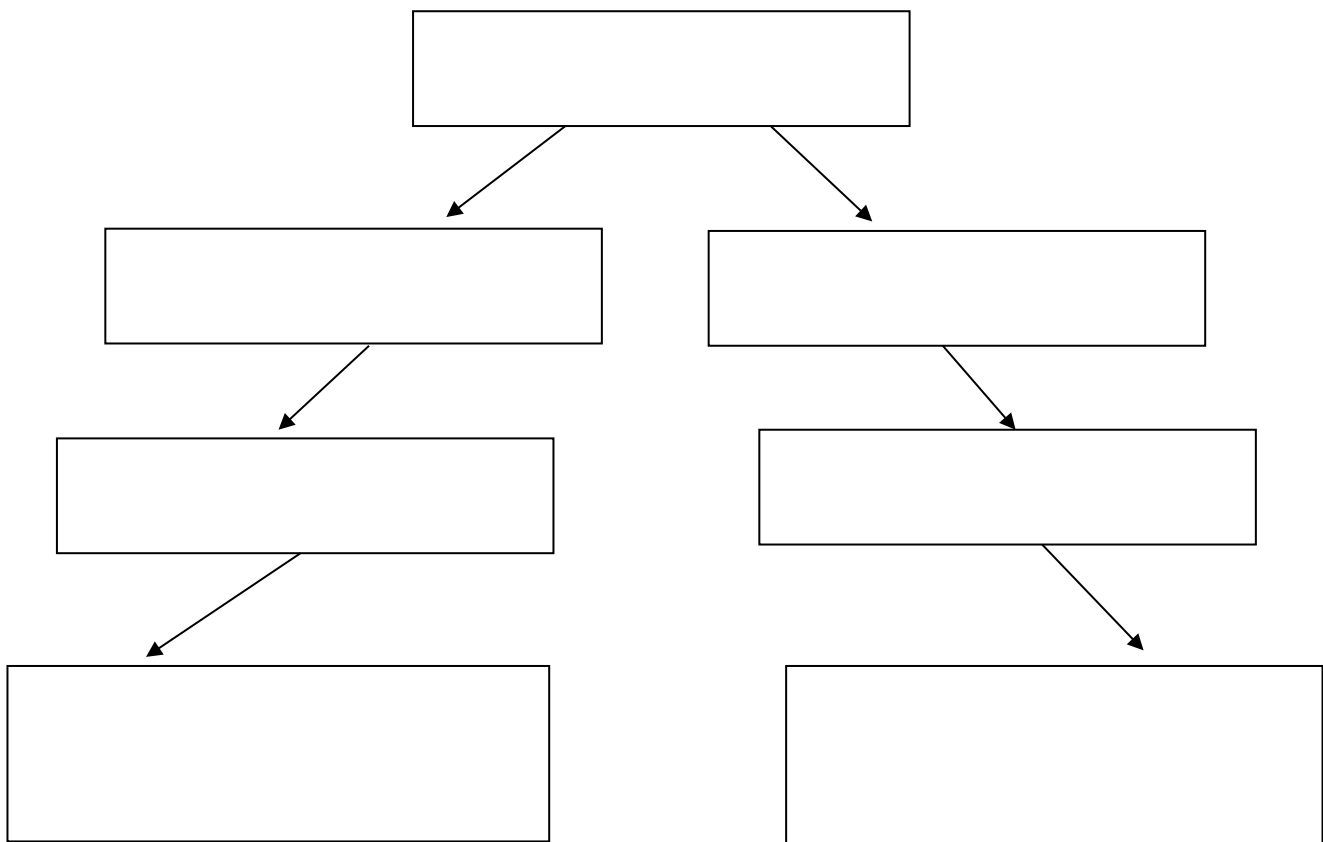
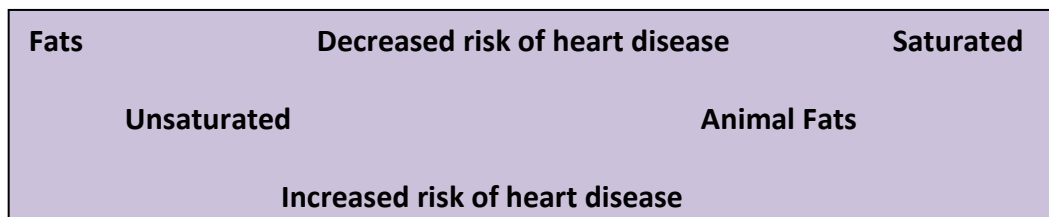
23. What is the difference between glycogen and glucose? How do they work together?
(pg. 125)

24. List 4 carbohydrates. (pg. 125)

25. What do we need fat for? (pg. 126)

26. List 4 sources of fat. (pg. 126)

27. Complete the following flow chart using the following words.



28. What is the difference between protein and carbohydrates? (pg. 127)

29. Why is protein important for our body? (pg. 127)

30. List 4 sources of protein. (pg. 127)

31. Why do children need more protein than adults? (pg. 127)

32. In what situations would your body use protein for energy? (pg. 127)

33. In your own words define metabolism? (pgs. 130 & 131)

34. Describe some of the differences in nutrient and energy requirements between a baby and a senior citizen? (pg. 130)

35. List 3 occupations in which you would expect workers to have high metabolisms? (pg. 131)

36. For each of the following situations, predict whether the person's metabolic rate will go up, down, or stay the same.

a) A construction worker resigns and takes on a rural mail route.

b) A mother stops breast feeding.

c) A high school gymnast begins training for the provincial finals

d) A weightlifter takes time off from training

Chapter 8: Fossil Fuels

Define the following:

Fossil Fuels

Coal

Natural gas

Crude Oil

Petroleum

Reservoir rock

Seismic survey

Pump jack

Lift pump

Distillation

Combustion

Hydrocarbons

Non-renewable

Acid deposition

Global climate change

1. List 3 fossil fuels. (pg. 137)

2. Sketch and LABEL four steps in the formation of coal. (pg. 136)

3. Describe two ways in which coal is mined. (pg. 138)

4. List 3 uses for each of the fossil fuels.(pg. 137)

5. Explain how the various components of crude oil can be separated. (pg. 143)

6. Sketch how oil, water and natural gas arrange themselves under the Earth's surface.
(pg. 140)

7. Explain how shockwaves are used to find oil.(pg. 140)
8. How are pumps used to bring oil to the surface?(pg. 141)
9. List 3 technologies that use combustion to convert the energy stored in fossil fuels into other forms of energy. (pg. 138-143)
10. Copy the word and chemical equations for the combustion of propane. $C_3H_8(g)$
Label the input energy, converter, and output energy. (pg. 144)
11. Identify the two gases produced by both combustion and cellular respiration. (pg. 146)
12. How else is cellular respiration similar to fossil fuel combustion? (pg. 146)

13. Why are fossil fuels said to be non-renewable? (pg. 147)

14. List 3 risks associated with our use of fossil fuels.(pg. 147 & 149)

15. Use a labeled diagram or flowchart to explain how an increase in greenhouse gas might make Earth warmer. (pg. 149)

16. Why is global climate change a concern? (pg. 149)